Visual cross-check of resting-state clustering

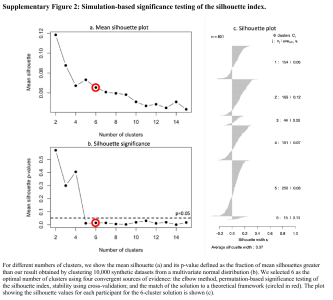
FOR2107 compared with Tozzi et al.

Same layout, different information content: the paper combines rest and task-derived circuit scores; our panels use the 14 resting-state edges currently available.

Preliminary comparison · five- and six-cluster descriptive solutions · 1 September 2026

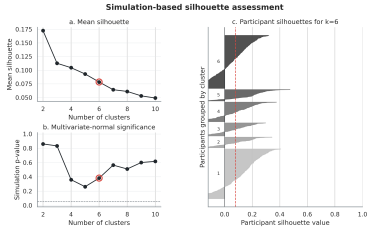
Supplementary Figure 2: how convincing is the separation?

Tozzi et al.



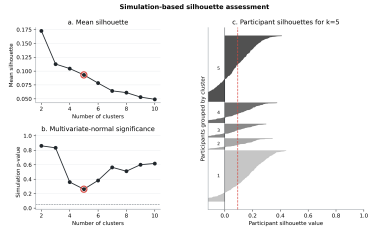
Original supplementary panel; six clusters circled in red.

FOR2107 · k = 6



14 resting-state edges; k=6 marked as the requested comparison.

FOR2107 · k = 5

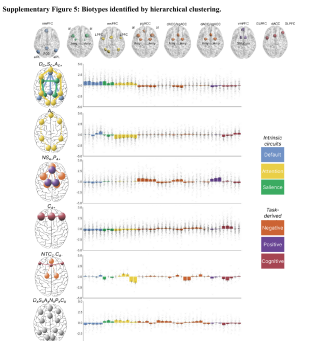


Same participants and features; hierarchy cut at five groups.

Reading this carefully: our permutation panel is significant because independent feature shuffling destroys real covariance. The covariance-preserving multinormal test is not significant for either requested solution.

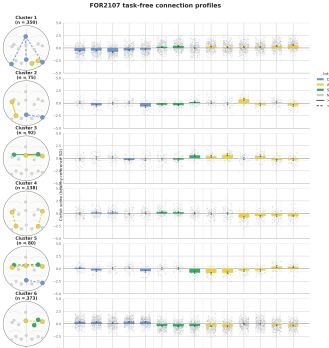
Supplementary Figure 5: what do the cluster profiles look like?

Tozzi et al.



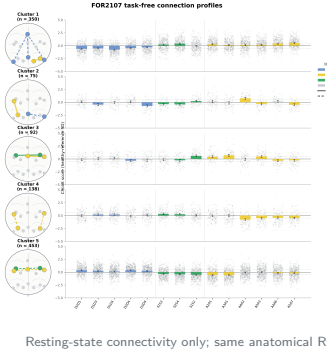
Original multimodal profiles: rest, task connectivity and task activity.

FOR2107 · k = 6



Resting-state connectivity only; gray bars are not FDR-significant.

FOR2107 · k = 5



Resting-state connectivity only; same anatomical ROI geometry.

What this comparison supports—and what it does not

What is already comparable

- **Correlation-distance, average-linkage hierarchy.**
- **Healthy-reference standardized circuit scores.**
- **Elbow, silhouette, permutation, resampling and split-half diagnostics.**
- **Anatomically matched Tozzi ROI definitions for the 14 rest edges.**

What remains unresolved

- **No evidence that k=5 or k=6 beats a covariance-preserving null.**
- **No task-derived activity/connectivity features yet.**
- **No Go/NoGo equivalent and no independent clinical replication.**
- **Cluster profiles are descriptive patterns, not validated biotypes.**

Current assessment: the figures reproduce the visual analysis framework, but the resting-only data do not yet justify selecting a biological cluster count.

Original paper panels are reproduced from the local supplementary PDF for scientific discussion; current panels are generated from repository outputs.